Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/30/2014
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Project Title:	Hypogravity induces vascular atrophy leading to bone loss		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Space Biology-FSB-Animal Physiology: NNH09ZTT003N
Start Date:	02/01/2010	**End Date:**	01/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:	Shuttle		
Flight Assignment:	STS-135 NOTE: Extended to 1/31/2014 per NSSC/A. Chu-ARC (Ed., 2/21/2013)		
Key Personnel Changes/Previous PI:	Dr. N. Patrick McCabe, the senior research investigator on the project, left the research program in mid-2012 and is no longer working on the project. Accordingly, readjustments of time management plans for existing research personnel needed to be enacted to complete the remaining tasks in this project. To these efforts Dr. Caroline Androjna provided daily supervision of a senior undergraduate Honors Thesis student, Mark Biro, recruited from Cleveland State University (Cleveland, OH). Mr. Biro's Honors Thesis work focused on completing the 3D micro-CT analysis of the barium-gelatin in-filled blood vessels in a 5 mm high mid-diaphyseal region-of-interest (ROI) in the mid-calf of rat lower hind limbs. The quantitative findings for perfused vascular volumes within 3 sub-regions representing blood vessels associated with the two bones in this 5 mm high ROI (tibia and fibula) are provided in Part A of this NCE progress report.		
COI Name (Institution):	Parsons-Wingerter, Patricia (NASA Glenn Research Center)		
Grant/Contract No.:	NNX10AE40G		
Performance Goal No.:			

Performance Goal Text:

Task Description:

Hypogravity leads to a loss of ground reaction forces transmitted through the lower appendicular and axial skeleton. Loss of weight bearing (WB) loads leads to a state of progressive osteopenia in these skeletal sites. Hypogravity mediated unloading reduces mechanical compression forces resulting in decreased local bone strains and reductions in hydrodynamic forces within bone tissue. In addition to this mechanism, we hypothesize hypogravity mediates reductions in the vascular supply serving these bone sites that contribute to hypogravity induced osteopenia and may help explain the prolonged time to recover bone mineral density (BMD) and bone strength upon return to Earth.
Vasculature associated with bone tissue will be assessed using 3D micro-angiography imaging, which uses contrast agents to fill blood vessels including those in bone followed by micro-CT imaging and histomorphometry. Using micro-angiography, we have developed the tissue and image processing means to assess the bone-associated vascular volumes in the hind limbs of rats. Using the hind limb unloaded (HLU) analog, a time-dependent loss of vascular volume in the lower legs will be monitored and compared to the temporal pattern of progressive bone loss at these skeletal sites. Blood flow and vascular volumes will be assessed in these rat groups and a subsequent effect of these treatments on bone volume, BMD, and vascular morphology in the tibiae and fibulae will be determined.

A cause & effect relationship between blood supply and osteopenia will need to be investigated in a future study by directly modulating blood supply to the hind limbs of rats. The blood supply to the hind limbs of WB rats could be reduced using unilateral femoral artery ligation and treatment with a VEGF inhibitor. The blood supply to the hind limbs of HLU rats could be enhanced using a hydrogel-based timed release of VEGF in proximity to the tibia and fibula. In addition, a loss of bone-associated blood supply could hamper a proper angiogenic response thereby delaying fracture healing. In this case, after HLU exposure, fibulae would be fractured and the extent of repair would be monitored by micro-CT imaging, followed by vascular volume assessments at the fracture callus site using 3D micro-angiography. Vascular volume, bone volume, BMD and vascular morphology within the fracture callus would be compared between HLU and WB rats.

This proposal frames a novel hypothesis that could contribute new knowledge impacting on both NASA's Risk of Accelerated Osteoporosis and its risk of Impaired Fracture Healing. If hypogravity unloading leads to an atrophy of bone-associated vasculature, then a negative impact could result contributing to the progressive osteopenia observed during space flight. A reduced bone vascular supply will hamper nutrient and waste product exchange rates in cortical bone thereby stressing osteocytes metabolically, and quite possibly leading to some osteocyte death and localized osteonecrosis. Given the slow remodeling rate of cortical bone, such a possibility could manifest in long term skeletal health complications. First, the presence of any localized osteonecrosis within cortical bone may elevate the risk of fracture upon return to Earth after a long space mission. Second, the remodeling & revascularization of cortical bone in skeletally mature adults takes a long time, and this slow process may help explain why the recovery rates for BMD and bone strength are slow in astronauts returning from long duration space missions. A slow recovery of bone-associated vascular supply may also compound into long term health effects to astronauts in their later years of life (e.g., peripheral artery disease).

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Our current knowledge base suggests that bone healing is impaired during and after an extended exposure to simulated weightlessness. This study frames a novel hypothesis that states that hypogravity mediated unloading leads to a progressive atrophy of the vasculature supplying lower leg bones. In turn, this regression in bone-associated blood supply and vascular bed volume will have a negative effect on bone tissues resulting in additional and progressive osteopenia beyond that attributed to reductions in local bone strain. Overall results from this investigation demonstrate that a direct correlation exists between vascular regression and deficient bone healing in the HLU space flight analog. These findings manifest potential relevance to NASA’s existing manned space missions as our discovery that vascular regression directly correlates with bone tissue regression in the HLU analog validates a novel hypothesis: hypogravity mediates reductions in vascular supplies serving lower leg bone sites that contribute to hypogravity induced osteopenia, and helps explain the prolonged BMD and bone strength recovery times upon return to Earth. Moreover, this hypothesis provides an additional mechanistic explanation for why deficient healing in lower leg bones exists in the HLU analog. These pre-clinical ground based studies may also exhibit importance regarding the underlying causes of impaired bone healing in patients experiencing paralysis, immobility, or extended bed rest here on Earth. Finally, this work may also exhibit relevance to other pathologies resulting in the poor healing of recalcitrant bone fractures here on Earth.

This final progress report represents a near completion of the work involved in the NNX10AE40G study: “Hypogravity induces vascular atrophy leading to bone loss”. A No-Cost-Extension (NCE) was requested for 2013-2014 because the senior research associate involved in this project (Dr. N. Patrick McCabe) left the Midura Lab in 2012 and readjustments of time management plans for existing research personnel needed to be enacted to complete the remaining tasks in this project. To these efforts Dr. Caroline Androjna provided daily supervision of a senior undergraduate Honors Thesis student, Mark Biro, recruited from Cleveland State University (Cleveland, OH). Mr. Biro’s Honors Thesis work focused on completing the 3D micro-CT analysis of the barium-gelatin in-filled blood vessels in a 5 mm high mid-diaphyseal region-of-interest (ROI) in the mid-calf of rat lower hind limbs. The quantitative findings for perfused vascular volumes within 3 sub-regions representing blood vessels associated with the two bones in this 5 mm high ROI (tibia and fibula) are provided in Part A of this NCE progress report.
This NCE progress report also includes an update on the mouse specimens flown aboard the STS-135 space shuttle mission (CBTM-03 payload), which was a supplement to NNX10AE40G. The relatedness of this spaceflight study compared with the ground based rat study should allow the two to be analyzed similarly and compared to one another. Part B of this NCE progress report will cover our latest progress with the STS-135 samples with specific emphasis on the immunohistochemical staining done on the tissue sections utilizing fluorescently tagged primary antibodies alpha-Smooth Muscle Actin (aSMA) and Isolectin B4 (IB4). Analysis of 2D vasculature morphometrics to quantify vessel size and vessel number in several regions of interest throughout the mid-calf tissues of the left hind limbs from untreated mice will be provided in this report. Part C of this NCE progress report will identify the remaining tasks that need to be completed in order to achieve complete datasets necessary for manuscript submission for peer-reviewed publication of our findings.

A. Ground based rat studies

Task Progress:

1) All animal based work has been completed, including all vascular perfusions and tissue retrieval from specimens. All micro-CT imaging and volumetric sample renderings have been completed. All tissue specimens have been processed for paraffin embedding and tissue cross sections have been completed. Mounted tissue sections have been imaged at 10X magnification in order to identify barium-perfused blood vessels.

2) These tissue sections are being used for immunohistochemical analysis and light microscopy. Barium-perfused vessels exhibit enough contrast with the tissue to be easily identified and quantitatively scored. As such, tissue sections of perfused animals were imaged using light microscopy and full tissue section images were compiled. The presence of barium particles in these perfused vessels was confirmed by sodium rhodizonate histochemistry [AJ Chaplin and ELT Turner. Observations on the histochemistry of barium. Histochemistry 79:111-116, 1983]. Some of these tissue sections were immunostained for alpha-Smooth Muscle Actin (aSMA) antibody and imaged by epifluorescent microscopy. The aSMA staining corroborated that these barium filled structures were indeed blood vessels ensuring a minimal number of false positive counts. These barium filled tissue sections are awaiting a completion of 2D analysis using ImagePro software. Such analysis will allow for 2D vessel metrics to be quantified including vessel size and vessel numbers. We have recruited an undergraduate intern who will start in May 2014 to complete these final morphometric assessments.

3) 3D micro-CT imaging of perfused rat specimens was completed and the raw data has undergone 3D morphometric analysis. These analyses have been validated for consistency and calibration thereby assuring a high level of precision and rigor. We can now compare our 3D datasets with existing literature values for 2D vascular morphometrics. In this regard, our assessment of the diameter of the great saphenous vein using these 3D datasets was calculated to be less than 10% different from literature values for this vein's diameter at the same position along the lower leg. 3D volumes of perfused rats were aligned to each animal's baseline scan due to each animal's anatomically unique characteristics.

4) 3D volumes of perfused hind limbs were cropped to cover a 5-mm vertical region of interest (ROI) within the middle of the lower hind limb. These crops were then divided into 3 additional regions of interest classified as anterior, central, and posterior based on the anatomical position in the hind limb. Once the vessels and bone had been aligned and cropped, the bones were masked out of the image this prevents the bone from interfering with the analysis of vasculature.

5) These 5 mm z-stack crops were then divided into 3 additional sub-ROIs classified as anterior, central, and posterior based on the anatomical position in the hind limb. Once the vessels and bone had been aligned and cropped, the bones were masked out of the image for an analysis of vasculature associated with these bones(vessels within 0.6 mm of the periosteal surface, within the bone tissue proper, and within the internal marrow space). This cropping analysis generated 5 distinct ROI sub-volumes: (1) Anterior soft tissues, (2) Central soft tissues, (3) Posterior soft tissues, (4) Tibia (anterior location), and (5) Fibula (posterior location).

6) Vascular volumes in these 5 mm high z-stacks were rendered for each of the 5 distinct ROI sub-volumes: (1) Anterior soft tissues, (2) Central soft tissues, (3) Posterior soft tissues, (4) Tibia (anterior location), and (5) Fibula (posterior location) and vessel morphometrics were extracted and calculated. Vascular volumes associated with the tibia and fibula were the specific focus of this investigation.

7) Data from the 5 distinct sub-volumes listed in #6 above needed to be precisely normalized as there were some slight differences between different sample volumes from like specimens within a given test group. For example, the 5 mm high sub-volume for all tibia specimens in this study ranged from 53-55 mm3 calculated total volume and the average value was calculated to be 54 mm3. Thus, all vascular volume data from all tibia specimens were normalized to a standard 54 mm3 volume. All vascular volume data from all fibula specimens were normalized to a standard 17 mm3 volume. Normalized tissue volume sizes for the remaining 3 sub-ROIs are: Anterior ROI, 300 mm3; Central ROI, 225 mm3; and Posterior ROI, 414 mm3

8) Data summary of the perfused vascular volumes associated with the 54 mm3 tibia region from baseline (start of study), 4 weeks or 8 weeks of cage activity (weight bearing, WB), and 4 weeks or 8 weeks of space flight analog (hind limb unloading, HLU): Four weeks of cage WB activity did not result in a significant change in tibia-associated vascular volume but 4 weeks of HLU exposure resulted in a 26% decline in tibia-associated vascular volume compared to the average baseline value ($p < 0.01$; one way ANOVA). Eight weeks of cage activity resulted in a significant 20% decline in tibia-associated vascular volume when compared to the average baseline value, and 8 weeks of HLU exposure resulted in a 34% decline in tibia-associated vascular volume compared to the average baseline value ($p < 0.01$; one way ANOVA). Eight weeks of HLU showed an 18% decline in tibia-associated vascular volume compared to the average 8 weeks WB value ($p < 0.05$; one way ANOVA). Four and eight weeks of HLU exhibited similar declines in tibia-associated vascular volume (not significantly different from each other; one way ANOVA) when compared to the average baseline value. Presently, we do not have a good explanation for why 8 weeks of cage WB activity would result in a decline in total vasculature associated with the tibial diaphysis from the average baseline value, but age-related changes in bone vasculature and/or reductions in cage activity levels would be obvious aspects that we would want to explore in future investigations. Overall, these data clearly indicate that the HLU analog generates reductions in the vascular volume associated with the tibia. These reductions in bone vasculature correlate with reductions in bone mass associated with the HLU model and we hypothesize that these two physiological events are functionally linked.

9) Data summary of the perfused vascular volumes associated with the 17 mm3 fibula region from baseline (start of study), 4 weeks or 8 weeks of cage activity (weight bearing, WB), and 4 weeks or 8 weeks of space flight analog (hind limb unloading, HLU): Four weeks of cage WB activity did not result in a significant change in fibula-associated vascular volume but 4 weeks of HLU exposure resulted in a 31% decline in fibula-associated vascular volume compared to the average baseline value ($p < 0.01$; one way ANOVA). Eight weeks of cage WB activity resulted in a significant 20% decline in tibia-associated vascular volume when compared to the average baseline value, and 8 weeks of HLU exposure resulted in a 25% decline in fibula-associated vascular volume compared to the average baseline value ($p < 0.01$; one way ANOVA). Eight weeks of HLU showed an 8% decline in fibula-associated vascular volume compared to the average 8 weeks WB value but this trend was not significant (one way ANOVA). Four and eight weeks of HLU exhibited similar declines in fibula-associated vascular volume (not significantly different from each other; one way ANOVA) when compared to the average baseline value. Presently, we do not have a good explanation for why 8 weeks of cage WB activity would result in a decline in total vasculature associated with the fibula diaphysis from the average baseline value, but age-related changes in bone vasculature and/or reductions in cage activity levels would be obvious aspects that we would want to explore in future investigations. Overall, these data clearly indicate that the HLU analog generates reductions in the vascular volume associated with the fibula. These reductions in bone vasculature correlate with reductions in fibula bone healing responses published for the HLU analog and we hypothesize that a reduction in

vasculature associated with the fibula could diminish bone healing responses and represent one physiological process linked to HLU's impairment of fracture healing.

B. Spaceflight Mouse Study

1) The genetic analysis of all STS-135 (CBTM-03) spaceflight and synchronous ground-based control samples has been completed. These specimens represent a similar mid-calf ROI as in the prior mentioned rat HLU study but only for the left hind limb of untreated control mice in this study. The bone sample was comprised of the tibia and its associated marrow tissue in the anterior ROI, while the muscle sample consisted of all of the skeletal muscle and soft connective tissues in the central and posterior ROIs of the mid-calf. These quantitative gene expression analyses were accomplished using Qiagen qRT-PCR arrays interrogating the activity of 88 genes related to endothelial cell function. The genetic expression data obtained from these arrays was analyzed using 7500 systems software and Qiagen supplied analysis software. Only ~6% and ~9% of the interrogated genes in the hind limb bone and muscle samples, respectively, exhibited >2-fold changes in normalized expression levels exposed to space flight in comparison to ground control counterparts. This relatively small set of changes in gene expression would suggest a relatively narrow and focused set of biological responses with respect to endothelial cell biology. Differences in gene expression outcomes between bone and muscle tissue compartments may be a reflection of the pronounced differences in the rates of muscle versus bone atrophy in the hind limb because the rate of sarcopenia is much faster than that of osteopenia and this will likely influence rates of vascular atrophy in these tissue compartments.

Bone samples: Matrix Metalloproteinase 1a (Mmp1a) is an important protease involved in vascular matrix remodeling and this collagenase activity allows for matrix removal by endothelial cells for subsequent vascular growth. An up regulation of Mmp1a might suggest an adaptive response by the vasculature as an attempt to compensate for the lower blood flow in hind limb bone/marrow tissue of space flight mice. Chemokine (C-X-C motif) receptor 5 (Cxcr5) is involved in regulating blood pressure, vasoconstriction, and angiotensin functions and a down regulation of Cxcr5 expression might suggest a decreased ability to regulate blood pressure, vasoconstriction, and angiotensin functions in hind limb bone/marrow tissue of space flight mice. Chemokine (C-C motif) ligand 2 (Ccl2) and Interleukin 1 beta (Il1b) are cytokine factors produced to stimulate cell growth & matrix production and are produced in response to metabolic stress and a down regulation of Ccl2 & Il1b expression might suggest a decreased ability for cell growth & matrix production as an adaptive response by the vasculature in hind limb bone/marrow tissue of space flight mice.

Muscle samples: B-cell leukemia/lymphoma 2-like 1 (Bcl2l1) is an anti-apoptosis factor produced by endothelial cells under stress or injury and an up regulation of Bcl2l1 may be an adaptive response to compensate for the metabolic stress resulting from lower blood flow in hind limb muscle tissue of space flight mice. Endothelin 2 (Edn2) is involved in regulating blood pressure, vasoconstriction, and angiotensin functions and an up regulation of Edn2 expression might suggest an attempt to readjust blood pressure, vasoconstriction, and angiotensin functions resulting from lower blood flow in hind limb muscle tissue of space flight mice. A disintegrin and metallopeptidase domain 17 (Adam17) and Matrix metalloproteinase 9 (Mmp9) are proteases involved in vascular matrix remodeling; Mmp9 is also expressed by leukocytes as a means to migrate across blood vessels and enter somatic tissues. Up regulation of Adam17 & Mmp9 may be adaptive responses by endothelial cells and/or leukocytes in response to lower blood flow in the hind limb muscle tissue of space flight mice. Vascular endothelial cadherin or Cadherin 5 (Cdh5) and Selectin, platelet ligand (Selplg) are factors involved in endothelium - leukocyte adhesion. Cadherin 5 is required for maintaining an endothelium barrier particularly that of newly formed vessels and P-selectin ligand is expressed by leukocytes enabling an interaction with P-selectin on the endothelial cell surface resulting in "rolling" of leukocytes on the endothelial cell surface allowing their stable adhesion and transmigration into somatic tissues. An up regulation of Cdh5 & Selplg expression might suggest the presence of endothelial cell activation & inflammation as a result of decreased blood flow in hind limb muscle tissues of space flight mice. Serine (or cysteine) peptidase inhibitor, clade E, member 1 (Serpine1) and Platelet factor 4 (Pf4; aka C-X-C motif chemokine 4) are factors that can enhance or decrease angiogenesis, respectively. Their biological effects depend upon whether vascular growth or regression stages predominant locally. Up regulation of Serpine1 & Pf4 expression might suggest an active state of vascular remodeling as an adaptive response by the vasculature to lower blood flow in hind limb muscle tissues of space flight mice.

2) Additional specimens for an HLU ground control study were obtained from Dr. Bouxseim's group at Harvard University. These samples included both HLU and WB control C57Bl/6 female mice (identical to those flown on STS-135). Specimens in the HLU group were exposed to HLU for the same duration as the space flight duration of the STS-135 mission (12.5 days). The control mice in this study were the same age as the HLU experimental mice and all animals were treated similarly to those from the STS-135 mission. The mid-calf ROI of each animal's left hind limb was utilized for this control study and, given the small size of the C57Bl/6 female mice used, the bone and skeletal muscle tissues were not separated. The proximal end was placed in RNAlater at 4° C to preserve RNA for genetic analysis of factors associated with endothelial cell alterations. The distal end was placed in 10% neutral-buffered formalin and stored at 4°C. The same analysis of gene expression was done using Qiagen gene arrays. This genetic data was also analyzed statistically using 7500 system software and Qiagen supplied analysis software. The distal ends were decalcified and embedded in tissue freezing medium and were cryo-sectioned for immunohistochemical analysis. The genetic analysis of the spaceflight and Harvard studies was divided into 4 unique groups; spaceflight ground control, spaceflight, Harvard ground control, and Harvard HLU. These groups were designated to allow for comparison of control groups as well as comparison of the spaceflight animals with the HLU mice. These findings are of great importance as hind limb unloading is a NASA approved model for spaceflight studies. Unexpectedly, when comparing HLU to WB mice within just the Harvard cohort, no gene expression patterns met our statistical cutoffs for the STS-135 flight and ground based control groups ($p<0.05$; fold change >2). Expressed as a ratio of HLU/WB target/housekeeping gene expression, three genes were relatively close to the cutoff threshold: Il1b showed a fold change of -2.14 ($p = 0.057$); Il11 showed a fold change of +1.79 ($p = 0.086$); and Cpb2 showed a fold change of +1.82 ($p = 0.065$).

3) It was determined that the von Willebrand factor (vWF) staining was too variable to be of use in the histomorphometric analysis of vasculature. As such it was found that isolectin B4 (IB4) gave more consistent and useful staining, especially of microvasculature (vessel diameters less than 50 microns). The immunohistochemical staining done on the tissue sections utilized fluorescently tagged primary antibody for alpha-Smooth Muscle Actin (aSMA) or Isolectin B4 (IB4). The tissue sections were stained along with DAPI which fluorescently labeled the nuclei. The tissue sections were then imaged on a fluorescent microscope and full-sized tissue section images were compiled. These immunohistochemical sections have undergone visual validation and preliminary morphometric analysis using ImagePro software. Analysis of vasculature will quantify vessel size and vessel number in several regions of interest throughout the tissue. Preliminary vascular morphometrics from STS-135 specimens are currently being recovered.

	C. Task progress that still remains to be done for publication purposes The 2D analysis of the tissue sections from the STS-135 arm of this investigation is currently being completed using a custom Image Pro software analysis built by a subcontractor for our study (Image IQ). I anticipate that it will take my research team another month to complete the entire work-plan for the STS-135 arm of NNX10AE40G. The 2D vascular morphometric analysis of vascular perfusions carried out in the ground based rat studies will be completed within the next 3 months by a summer undergraduate intern recruited to complete this final task for the rat HLU arm of NNX10AE40G.
Bibliography Type:	Description: (Last Updated: 03/01/2017)
Abstracts for Journals and Proceedings	Androjna C, McCabe NP, Cavanagh PR, Globus RK, Hill E, Ke HZ, Warden S, Burr D, Hill-Gallant K, Midura RJ. "Bone Healing is Altered by Space Flight and Unloading." 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3027.pdf , Feb-2014
Articles in Peer-reviewed Journals	McCabe NP, Androjna C, Hill E, Globus RK, Midura RJ. "Simulated microgravity alters the expression of key genes involved in fracture healing." Acta Astronautica. 2013 Nov;92(1):65-72. http://dx.doi.org/10.1016/j.actaastro.2012.04.016 , Nov-2013